An adaptive-step Adams-Bashforth-Moulton optimizer with Gaussian process surrogate for the lightweight design of automotive structural components

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Abstract

The Adams-Bashforth-Moulton optimizer (ABMO) is a recent mathematics-based metaheuristic that models the movement of candidate solutions with the predictor corrector scheme of multistep integration. Its step size follows a fixed schedule, and its authors identify adaptive step control as the main open question of their work. This paper shows that the predictor corrector stages of ABMO are collinear, so that the local error estimate of the integration method cannot be transferred to the optimizer, and proposes AS-ABMO, which combines an individual-aware predictor with a success-rate step control. A faithful open re-implementation of ABMO is provided and its published results are reproduced. An ablation study shows that neither component alone is sufficient, whereas their combination improves the base algorithm on eight of twelve CEC2022 functions in ten dimensions and on five of twelve in twenty dimensions with no significant loss and no measurable cost. Under an identical evaluation budget AS-ABMO ranks second among eight optimizers, behind L-SHADE and ahead of RUN, GWO, WOA, HHO and SCA. On seven constrained design problems the medians of both ABMO variants coincide with the exact optima obtained by multistart sequential quadratic programming. The optimizer is finally coupled with a Gaussian process surrogate of a finite element model for the mass minimization of [component], where [result].

Keywords: metaheuristic optimization; Adams-Bashforth-Moulton optimizer; adaptive step size; constrained engineering design; surrogate-assisted optimization; lightweight design

1 Introduction

Mass reduction of load-bearing components is a central objective in automotive and aerospace design, since every kilogram removed from a vehicle structure lowers fuel or battery consumption over the service life without changing the powertrain. The design task is a constrained optimization problem: geometric parameters such as wall thicknesses, rib positions and fillet radii must be selected so that the mass is minimized while stress, deflection, buckling or fatigue limits are respected. The objective and the constraints are usually evaluated by finite element analysis (FEA), are non-convex, and may be discontinuous or non-differentiable in the design variables. For this class of problems metaheuristic optimizers have become the standard tool, and a substantial body of work has applied them to structural components such as suspension arms, brackets, wing ribs and hinges [1, 2, 3, 4, 5, 6, 7].

Metaheuristics are commonly grouped by their source of inspiration into evolutionary, swarm-based, human-based, physics-based and mathematics-based methods. The last group derives its search operators from numerical procedures rather than from a metaphor. Examples are the gradient-based optimizer [8], the arithmetic optimization algorithm [9], the Runge Kutta optimizer (RUN)

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[10], the quadratic interpolation optimizer (QIO) [11], the Newton Raphson based optimizer (NRBO) [12] and, most recently, the Adams-Bashforth-Moulton optimizer (ABMO) [13]. Mathematics-based methods are attractive for engineering use because their update rules are transparent and because they are not exposed to the criticism directed at metaphor driven algorithms [14]. At the same time, the no free lunch theorem [15] implies that no single optimizer dominates on all problems, and practical experience shows that a base algorithm usually benefits from targeted modifications before it is applied to a specific design domain. Chaotic maps, opposition-based learning, adaptive step schedules and hybridization with local search have all been used for this purpose; the chaotic Runge Kutta algorithm [16] is a representative example of such a modification of a mathematics-based optimizer for constrained engineering design.

ABMO models the movement of a candidate solution with the four-step Adams-Bashforth predictor and the three-step Adams-Moulton corrector that are used for the numerical integration of ordinary differential equations. Its authors report a first rank among twelve algorithms on the CEC2017 and CEC2022 suites and competitive results on nine constrained design problems [13]. Three aspects of the original work motivate the present study. First, no source code was released with the paper, so that independent verification and reuse depend on a faithful re-implementation. Second, the authors explicitly list adaptive control of the step size as the first open research question of their work: the step h and the balance factor E follow fixed schedules in the number of iterations and do not respond to the progress of the search. Third, an inspection of the update equations shows that the four predictor stages and the corrector stage are, up to element-wise random scaling, multiples of a single population vector. This structural property, which is analysed in Section 2, means that the classical error estimate of predictor corrector integration cannot be transferred to the optimizer and that any adaptive step control has to be based on feedback from the fitness values instead.

In this paper we propose the adaptive-step ABMO (AS-ABMO), a two-component modification of the base algorithm, and apply it to the lightweight design of an automotive structural component with a Gaussian process surrogate of the finite element model. The contributions are as follows.

1. A faithful open re-implementation of ABMO is provided and its published results are reproduced on the CEC2022 suite and on the classical constrained design problems; discrepancies with the original report are documented.

2. The predictor corrector stages of ABMO are shown to be collinear, which explains why a Milne-type local error estimate carries no information about the search and motivates a feedback controller.

3. AS-ABMO adds an individual-aware predictor, which lets each candidate solution contribute its own direction to the prediction, and a success-rate step control that scales the step according to the fraction of accepted moves. An ablation study shows that neither component alone is sufficient and that their combination improves the base algorithm on eight of twelve CEC2022 functions in ten dimensions and on five of twelve in twenty dimensions, without a single significant loss.

4. Under an identical evaluation budget AS-ABMO ranks second among eight optimizers on CEC2022, behind L-SHADE and ahead of the mathematics-based RUN and of the widely used GWO, WOA, HHO and SCA. On seven constrained design problems the medians of AS-ABMO coincide with the exact optima obtained by multistart sequential quadratic programming; a feasibility rule is used for all algorithms so that the comparison does not depend on penalty coefficients.

5. The optimizer is coupled with a Gaussian process surrogate of a finite element model and applied to the mass minimization of [COMPONENT, to be completed], with FEA verification of the optimized design.

The remainder of the paper is organized as follows. Section 2 recalls ABMO and presents the collinearity analysis. Section 3 describes AS-ABMO. Section 4 reports the reproduction study, the ablation, the parameter sensitivity and the comparison on CEC2022. Section 5 treats the constrained design problems, Section 6 the component application, and Section 7 discusses the results and their implications for design practice. Section 8 concludes.

2 The Adams-Bashforth-Moulton optimizer

2.1 Search mechanism

ABMO operates on a population of N candidate solutions $\mathbf{X}_i \in \mathbb{R}^D$, $i = 1, \ldots, N$, initialized uniformly within the bounds $[\mathbf{l}, \mathbf{u}]$ of the problem. Let $\mathbf{X}_{\text{best}}$ denote the best solution found so far and $\mathbf{X}_{\text{mean}}$ the arithmetic mean of the population at iteration t of T. The original formulation [13] defines a balance factor and a step size,

$$E_t = \exp\big(k\,(t/T)^2\big)\,, \qquad h = \exp(-(1-2t/T))\; r_1, \tag{1}$$

with $k = -10$ and $r_1 \sim U(0,1)$, and builds the four stages of the Adams-Bashforth predictor from the population direction $\mathbf{f}_1 = \mathbf{X}_{\text{mean}} - \mathbf{X}_{\text{best}}$:

$$\mathbf{f}_{j+1} = \mathbf{f}_1 + \tfrac{h}{2}\,\mathbf{r} \odot \mathbf{f}_j, \qquad j = 1, 2, 3, \tag{2}$$

where $\mathbf{r} \sim U(0,1)^D$ and $\odot$ is the element-wise product. The predicted position and the corrected position are

$$\mathbf{X}_{\text{pred}} = \mathbf{X}_{\text{best}} + \tfrac{h}{24}\,(55\,\mathbf{f}_1 - 59\,\mathbf{f}_2 + 37\,\mathbf{f}_3 - 9\,\mathbf{f}_4)\,, \tag{3}$$

$$\mathbf{f}_5 = \tfrac{1}{2}\,(\mathbf{X}_{\text{mean}} - w\,\mathbf{X}_{\text{best}})\,, \qquad w = \exp\big(-\tfrac{t}{2T}\,r_2\big)\,, \tag{4}$$

$$\mathbf{S} = \mathbf{X}_{\text{pred}} + \tfrac{h}{24}\,(19\,\mathbf{f}_1 - 5\,\mathbf{f}_2 + \mathbf{f}_3 + 9\,\mathbf{f}_5)\,. \tag{5}$$

If a uniform random number does not exceed E_t the algorithm is in its exploration phase and a random member $\mathbf{X}_q$ of the population is used to perturb the corrected position,

$$\mathbf{X}_{\text{new}} = \mathbf{S} + (\mathbf{X}_{\text{mean}} - \mathbf{X}_q) \odot \cos\big(\tfrac{\pi}{2}\mathbf{r}_3\big) + (\mathbf{X}_{\text{best}} - \mathbf{X}_{\text{mean}}) \odot \cos\big(\tfrac{\pi}{2}\mathbf{r}_4\big)\,; \tag{6}$$

otherwise the exploitation phase blends the corrected position with the incumbent and adds a sinusoidal perturbation,

$$\mathbf{X}_{\text{new}} = (1-d)\,\mathbf{X}_{\text{best}} + d\,\mathbf{S} + (\mathbf{X}_{\text{best}} - \mathbf{X}_i) \odot \sin(2\pi\,\mathbf{r}_5)\,, \qquad d = \exp(t - T). \tag{7}$$

A greedy rule keeps $\mathbf{X}_{\text{new}}$ only if it improves on $\mathbf{X}_i$. Since $d = \exp(t-T)$ is numerically zero for all but the last few iterations, the exploitation phase is in practice a local search around $\mathbf{X}_{\text{best}}$ driven by the last term of Eq. (7). We keep this schedule unchanged: in a preliminary ablation a linear replacement $d = t/T$ was the worst of six configurations tested (Section 4.3).

2.2 Collinearity of the predictor corrector stages

In the integration of an ordinary differential equation the difference between the corrected and the predicted value estimates the local truncation error (Milne's device) and is the basis of automatic step size control. It is natural to ask whether the same quantity could control the step h in ABMO. Equation (2) shows that this is not the case. Writing $\mathbf{c} = \frac{h}{2}\mathbf{r}$, the stages are

$$\mathbf{f}_2 = (\mathbf{1} + \mathbf{c}) \odot \mathbf{f}_1, \quad \mathbf{f}_3 = (\mathbf{1} + \mathbf{c} + \mathbf{c}^{\odot 2}) \odot \mathbf{f}_1, \quad \mathbf{f}_4 = (\mathbf{1} + \mathbf{c} + \mathbf{c}^{\odot 2} + \mathbf{c}^{\odot 3}) \odot \mathbf{f}_1, \tag{8}$$

so that every stage is an element-wise rescaling of the same vector $\mathbf{f}_1$. The stages are therefore collinear in each coordinate; they do not sample the objective at distinct points and do not encode any curvature of the landscape, which is what the error estimate of a multistep integrator relies on. The corrector, Eq. (5), adds the term $\mathbf{f}_5$, which depends on the absolute positions $\mathbf{X}_{\text{mean}}$ and $\mathbf{X}_{\text{best}}$ rather than on their difference and therefore dominates $\mathbf{S} - \mathbf{X}_{\text{pred}}$ whenever the population is far from the origin. As a consequence the predictor corrector difference measures the random draws $\mathbf{r}$ and w and the location of the population, not the quality of the prediction. This is confirmed experimentally in Section 4.3, where a Milne-type controller based on $\|\mathbf{S} - \mathbf{X}_{\text{pred}}\|$ does not change the performance of the base algorithm. Two conclusions follow for the design of an adaptive variant: the step size has to be controlled by feedback from the accepted moves, and the prediction should be given access to information that differs between individuals.

2.3 Constraint handling and reproduction

For constrained problems the original work describes an adaptive penalty with an initial coefficient of 100 that is updated every 50 iterations from the feasible ratio of the population [13]. In our re-implementation this scheme did not reproduce the reported results: on the three-bar truss and the pressure vessel problems the population converged to strongly infeasible regions because the penalty remained small compared with the gain in the objective. We therefore use the feasibility rule of Deb [17] for ABMO and for every competitor: a feasible solution is preferred to an infeasible one, two feasible solutions are compared by objective value and two infeasible solutions by the sum of squared constraint violations. The rule has no parameters, which removes penalty tuning as a confounding factor in the comparison. With this rule the re-implemented ABMO attains the exact optima of the truss, spring, pressure vessel and welded beam problems (Section 5) and reproduces the CEC2022 results of the original paper within the variation expected between independent implementations (Section 4.2).

3 Proposed method: AS-ABMO

AS-ABMO modifies ABMO in two places and leaves every other operator, including the schedules of E_t, h, w and d, unchanged. The two modifications respond to the two conclusions of Section 2.2.

3.1 Individual-aware predictor

In ABMO all members of the population predict from the same direction $\mathbf{f}_1 = \mathbf{X}_{\text{mean}} - \mathbf{X}_{\text{best}}$, so that the predicted positions of all individuals lie on one line through $\mathbf{X}_{\text{best}}$ and differ only by the random scalings of Eq. (2). AS-ABMO lets each individual add its own offset from the population mean to the prediction direction,

$$\mathbf{f}_{1,i} = (\mathbf{X}_{\text{mean}} - \mathbf{X}_{\text{best}}) + \boldsymbol{\rho}_i \odot (\mathbf{X}_i - \mathbf{X}_{\text{mean}}), \qquad \boldsymbol{\rho}_i \sim U(0,1)^D, \tag{9}$$

and uses $\mathbf{f}_{1,i}$ in place of $\mathbf{f}_1$ in Eqs. (2)–(5). The predictor now targets a random point between the population mean and the individual instead of the mean itself. The expected direction is unchanged, since $\mathbb{E}[\boldsymbol{\rho}_i \odot (\mathbf{X}_i - \mathbf{X}_{\text{mean}})]$ averages to zero over the population, but the predicted positions are spread over the region occupied by the population. No additional function evaluation is required.

3.2 Success-rate step control

The step h of Eq. (1) is multiplied by a scale s_t that is updated once per iteration from the fraction of accepted moves. Let n_t be the number of individuals whose new position was accepted by the greedy rule at iteration t; in constrained problems only acceptances in which both the old and the new position are feasible are counted, so that improvements of the constraint violation do not inflate the step. The scale is updated as

$$s_{t+1} = \min\left\{s_{\max},\ \max\left\{s_{\max}^{-1},\ s_t \exp\left(\tfrac{n_t}{N} - p^*\right)\right\}\right\}, \qquad s_1 = 1, \tag{10}$$

and the effective step used in Eqs. (2)–(5) is $h_t = s_t\, h$. When more than a fraction p^* of the moves is accepted the step grows, when fewer are accepted it shrinks; the update is the multiplicative form of the classical one-fifth success rule of evolution strategies [18]. Two parameters are introduced, the target success rate p^* and the bound $s_{\max}$. The sensitivity study of Section 4.4 selects $p^* = 0.1$ and $s_{\max} = 10$, and the same values are used for all problems in this paper. Algorithm 1 and Fig. 1 summarize the method; the changes with respect to ABMO are marked.

Algorithm 1: AS-ABMO

Input: objective f, bounds $[\mathbf{l}, \mathbf{u}]$, N, T, $k = -10$, $p^* = 0.1$, $s_{\max} = 10$

Initialize $\mathbf{X}_i \sim U(\mathbf{l}, \mathbf{u})$, evaluate, set $\mathbf{X}_{\text{best}}$; $s_1 \leftarrow 1$;
for $t = 1$ **to** T **do**
    $E_t \leftarrow \exp(k(t/T)^2)$; $d \leftarrow \exp(t - T)$; $\mathbf{X}_{\text{mean}} \leftarrow \frac{1}{N}\sum_i \mathbf{X}_i$; $n_t \leftarrow 0$;
    for $i = 1$ **to** N **do**
        $\mathbf{f}_{1,i} \leftarrow (\mathbf{X}_{\text{mean}} - \mathbf{X}_{\text{best}}) + \boldsymbol{\rho}_i \odot (\mathbf{X}_i - \mathbf{X}_{\text{mean}})$; `// Eq. (9), new`
        $h \leftarrow s_t \exp(-(1 - 2t/T))\, r_1$; `// scaled step, new`
        compute $\mathbf{f}_2, \mathbf{f}_3, \mathbf{f}_4$, $\mathbf{X}_{\text{pred}}$, $\mathbf{f}_5$, $\mathbf{S}$ by Eqs. (2)–(5);
        if $r \le E_t$ **then**
            $\mathbf{X}_{\text{new}} \leftarrow$ Eq. (6)
        else
            $\mathbf{X}_{\text{new}} \leftarrow$ Eq. (7)
        clip $\mathbf{X}_{\text{new}}$ to $[\mathbf{l}, \mathbf{u}]$, evaluate;
        if $\mathbf{X}_{\text{new}}$ *is better than* $\mathbf{X}_i$ *(feasibility rule)* **then**
            if *both feasible* **then** $n_t \leftarrow n_t + 1$;
            $\mathbf{X}_i \leftarrow \mathbf{X}_{\text{new}}$; update $\mathbf{X}_{\text{best}}$;
    $s_{t+1} \leftarrow \text{clip}\big(s_t \exp(n_t/N - p^*),\ s_{\max}^{-1},\ s_{\max}\big)$; `// Eq. (10), new`

Output: $\mathbf{X}_{\text{best}}$

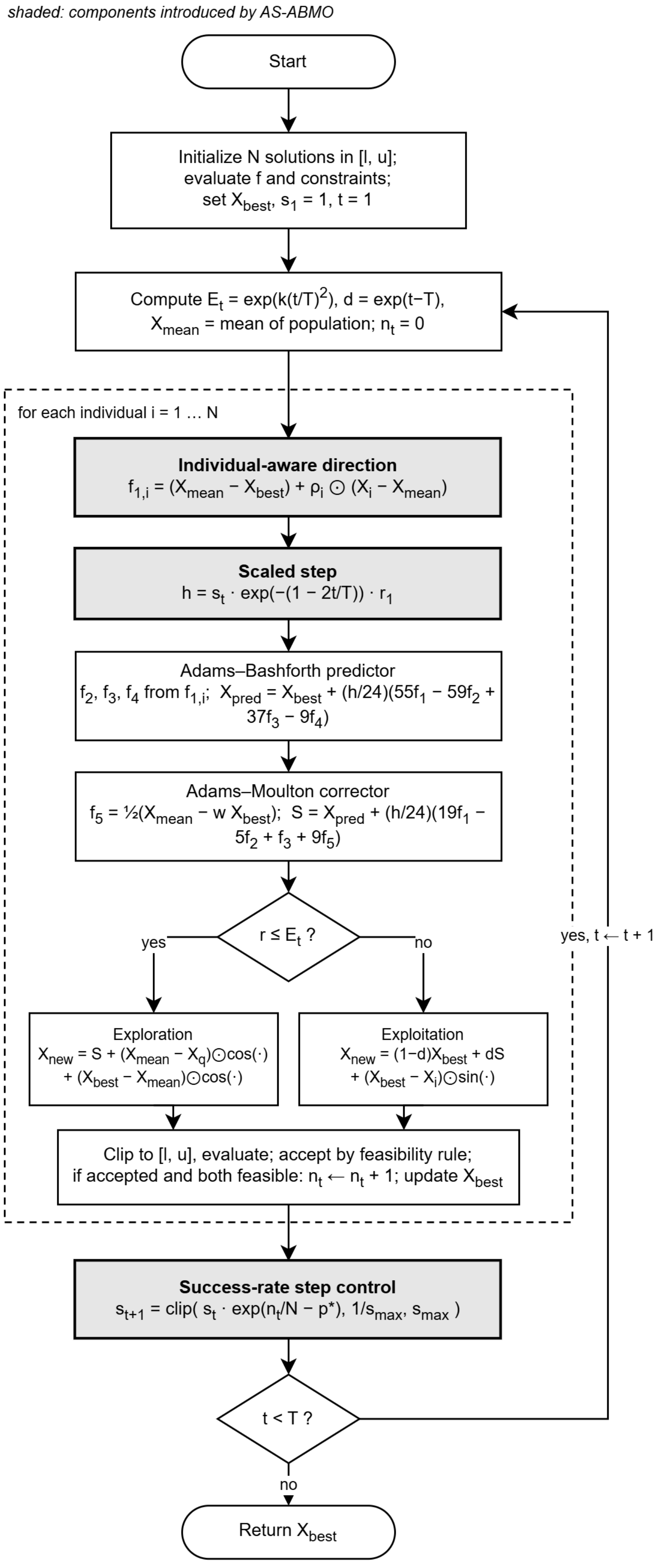

Figure 1: Flowchart of AS-ABMO. Shaded blocks are the components introduced in this paper; all other blocks are those of ABMO.

3.3 Computational complexity

Both modifications add $O(D)$ operations per individual and one scalar update per iteration, so the complexity of AS-ABMO remains $O(TND)$ as for ABMO; the number of objective evaluations is identical, $N(T+1)$. The measured run times in Section 4.7 confirm that the overhead is below the resolution of the timing.

4 Numerical experiments

4.1 Experimental setup

Figure 2 gives an overview of the study.

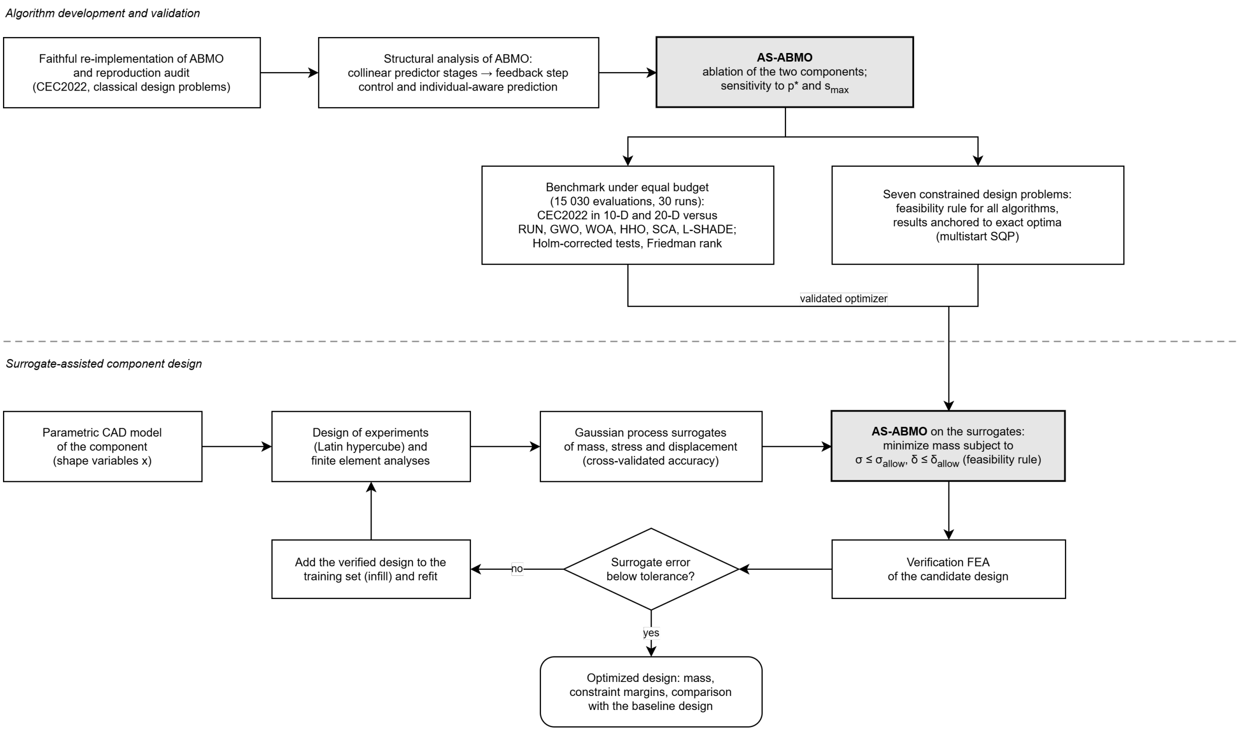

Figure 2: Workflow of the study: algorithm development and validation (top) and surrogate-assisted component design (bottom).

All experiments use the CEC2022 single-objective suite (twelve functions: F1 unimodal, F2–F5 basic multimodal, F6–F8 hybrid, F9–F12 composition) in $D = 10$ and $D = 20$ [19]. Following the original ABMO paper, the population size is $N = 30$ and the number of iterations $T = 500$, which corresponds to 15 030 function evaluations including initialization; every algorithm is given exactly this budget. Each configuration is run 30 times with fixed seeds. Competitors are the mathematics-based RUN [10], the widely used GWO [20], WOA [21], HHO [22] and SCA [23], and the CEC-competition algorithm L-SHADE [24], all with the parameter settings of their original papers as implemented in the `mealpy` library [25]. ABMO and AS-ABMO are implemented in Python by the authors; the code and all raw results are available in the repository listed in the data availability statement. Differences between AS-ABMO and each competitor are tested per function with the two-sided Mann–Whitney test at $\alpha = 0.05$ with Holm correction over the twelve functions, and summarized as wins, ties and losses (W/T/L) of AS-ABMO. Overall rankings use the Friedman mean rank of the mean final fitness. Experiments were run on a workstation with an Intel Core i9-14900K processor and 128 GB of memory under Python 3.11.

4.2 Reproduction of the base algorithm

Table 1 compares the mean final fitness of the re-implemented ABMO with the values reported in the original paper for $D = 10$. Eight of the twelve means are within $\pm10\%$ of the published values, and on F1, F3 and F5 the re-implementation reaches the global optimum where the published means are higher. The largest discrepancy is on the hybrid function F6, whose distribution over runs is heavy-tailed (minimum 3354 against a mean of 24 886); the published mean for this function is 3210. We attribute the remaining differences to the random number streams and to unreported implementation details, and we consider the base algorithm reproduced to the extent that a faithful re-implementation from the published equations allows. All subsequent comparisons use our implementation for both ABMO and AS-ABMO, so that the modifications are evaluated against the same code base.

Table 1: Reproduction of ABMO on CEC2022, $D = 10$, 30 runs: mean and standard deviation of the final fitness obtained with the re-implementation, and the mean reported in the original paper.

	Mean	Std	Reported		Mean	Std	Reported
F1	300.0	0.0	1.35E3	F7	2061.1	54.7	2.08E3
F2	423.9	50.8	4.06E2	F8	2285.8	162.6	2.26E3
F3	600.0	0.0	6.26E2	F9	2681.7	51.2	2.53E3
F4	876.6	32.4	8.36E2	F10	2879.9	406.4	2.63E3
F5	901.9	2.2	1.11E3	F11	2809.1	376.2	2.71E3
F6	24886	16520	3.21E3	F12	2895.1	51.3	2.87E3

4.3 Ablation study

Two ablations were carried out in $D = 10$. The first tested the step control alternatives discussed in Section 2.2 on the unmodified predictor: a Milne-type controller driven by $\|\mathbf{S} - \mathbf{X}_{\text{pred}}\|$, the success-rate controller of Eq. (10), and the linear schedule $d = t/T$ in Eq. (7), alone and in combination. The Milne controller did not change the Friedman rank of the base algorithm (3.88 against 3.83), the linear d schedule was the worst of the six configurations (4.83), and the success-rate controller was the best (1.54). The second ablation, reported in Table 2 and Fig. 3, isolates the two components of AS-ABMO. The individual-aware predictor alone (+IA) is not useful: it spreads the predicted positions but the fixed step schedule cannot exploit the added diversity, and its rank (3.29) is no better than that of ABMO (3.21). The success-rate controller alone (+AS) improves the rank to 1.96 but is significantly better than ABMO on only two functions after Holm correction. The combination (+IA+AS) reaches a rank of 1.54 and is significantly better than ABMO on eight of the twelve functions with no loss; on F6 the mean error is reduced by more than half. The two components are therefore complementary: the predictor supplies individual directions and the controller regulates how far they are followed.

Table 2: Ablation of the two components on CEC2022, $D = 10$, 30 runs: mean final fitness, Friedman mean rank, and wins/ties/losses against ABMO (Mann–Whitney, Holm-corrected, $\alpha = 0.05$).

	ABMO	+AS	+IA	+IA+AS
F1	300.00	300.00	300.00	300.00
F2	413.74	410.51	420.42	407.51
F3	600.00	600.00	600.00	600.00
F4	879.79	843.24	882.15	844.90
F5	902.15	900.99	902.23	900.22
F6	24123.9	14561.7	21043.6	11281.3
F7	2071.76	2052.07	2063.91	2030.70
F8	2314.48	2267.52	2265.09	2221.70
F9	2668.87	2595.64	2667.73	2603.91
F10	2913.87	2799.89	2915.06	2769.79
F11	2782.44	2782.65	2839.56	2836.58
F12	2892.23	2886.72	2899.33	2869.72
Friedman rank	3.21	1.96	3.29	1.54
W/T/L vs. ABMO	–	2/10/0	1/11/0	8/4/0

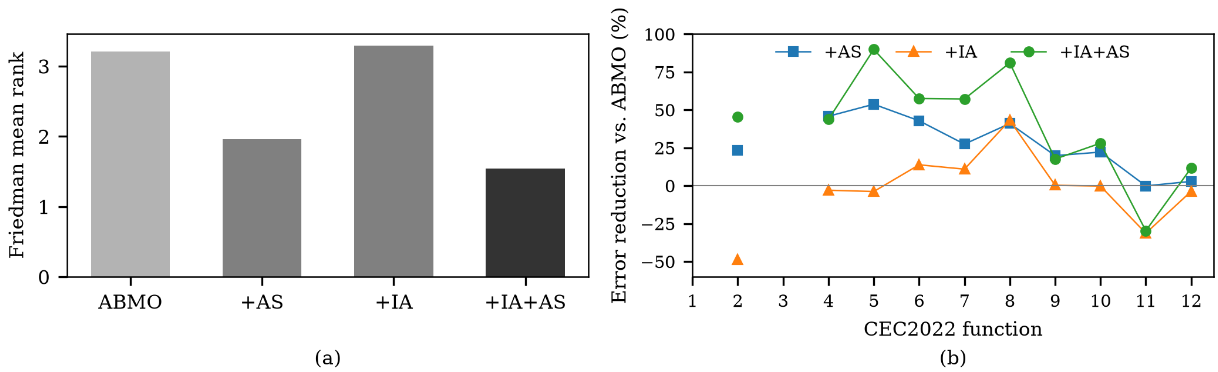

Figure 3: Ablation on CEC2022, $D = 10$: (a) Friedman mean rank of the four configurations; (b) reduction of the mean error $f - f^*$ relative to ABMO per function. F1 and F3 are solved to the global optimum by all variants and are omitted from (b).

4.4 Parameter sensitivity

The two parameters introduced by the controller were varied over $p^* \in \{0.1, 0.2, 0.3\}$ and $s_{\max} \in \{3, 10, 30\}$ with the full method in $D = 10$ (Table 3). A low target success rate is preferable: with $p^* = 0.1$ the step shrinks only when fewer than three of thirty moves are accepted, which keeps the search moving late in the run. The bound $s_{\max} = 10$ gives the best rank; $s_{\max} = 3$ restricts the controller too much and $s_{\max} = 30$ allows steps that are frequently clipped at the bounds. The selected pair $(0.1, 10)$ is used throughout the paper, including the constrained problems of Section 5, where a separate check showed no significant difference among $s_{\max} \in \{3, 10\}$.

Table 3: Sensitivity of AS-ABMO to the controller parameters on CEC2022, $D = 10$: Friedman mean rank over the twelve functions of the nine configurations (lower is better).

	$s_{\max} = 3$	$s_{\max} = 10$	$s_{\max} = 30$
$p^* = 0.1$	5.17	**2.75**	3.33
$p^* = 0.2$	5.33	4.42	4.50
$p^* = 0.3$	7.08	6.25	6.17

4.5 Comparison with other optimizers

Tables 5 and 6 report mean and standard deviation of the final fitness for the eight algorithms, and Table 4 summarizes the statistical comparison. In both dimensions L-SHADE is the best algorithm, which is expected for a method developed for the CEC competitions and equipped with parameter adaptation and population reduction. AS-ABMO ranks second (3.21 in $D = 10$, 2.33 in $D = 20$) and is ahead of RUN, the other mathematics-based competitor, of GWO, WOA, HHO and SCA, and of the base algorithm. Against ABMO it wins 8 functions in $D = 10$ and 5 in $D = 20$ without a loss; against RUN 8 and 11; against GWO 7 and 8 with two and one losses; and it loses to L-SHADE on 11 and 10 functions. The advantage over the base algorithm and over GWO and RUN increases with the dimension. Figure 4 shows mean convergence curves for F4, F6, F7 and F12: ABMO reaches a plateau within the first 50 to 100 iterations, whereas AS-ABMO continues to improve, in $D = 20$ typically until iteration 150 to 300. On F6 in $D = 20$ the modified algorithm descends more slowly in the first 100 iterations because the enlarged step keeps the population dispersed, and both variants end at the same level. Figure 5 shows the distributions of the final fitness; AS-ABMO has a narrower spread than ABMO on all eight panels and removes the outliers of ABMO on F12.

Table 4: Friedman mean rank of the mean final fitness on CEC2022, and wins/ties/losses of AS-ABMO against each competitor (Mann–Whitney, Holm-corrected over the twelve functions, $\alpha = 0.05$).

	$D = 10$		$D = 20$	
Algorithm	Rank	W/T/L	Rank	W/T/L
L-SHADE	1.00	0/1/11	1.08	0/2/10
AS-ABMO	3.21	–	2.33	–
GWO	4.33	7/5/0	4.00	8/3/1
RUN	4.50	8/4/0	4.92	11/0/1
ABMO	4.79	8/4/0	4.00	5/7/0
SCA	5.17	9/1/2	6.75	11/0/1
HHO	6.42	11/0/1	6.83	12/0/0
WOA	6.58	11/0/1	6.08	12/0/0

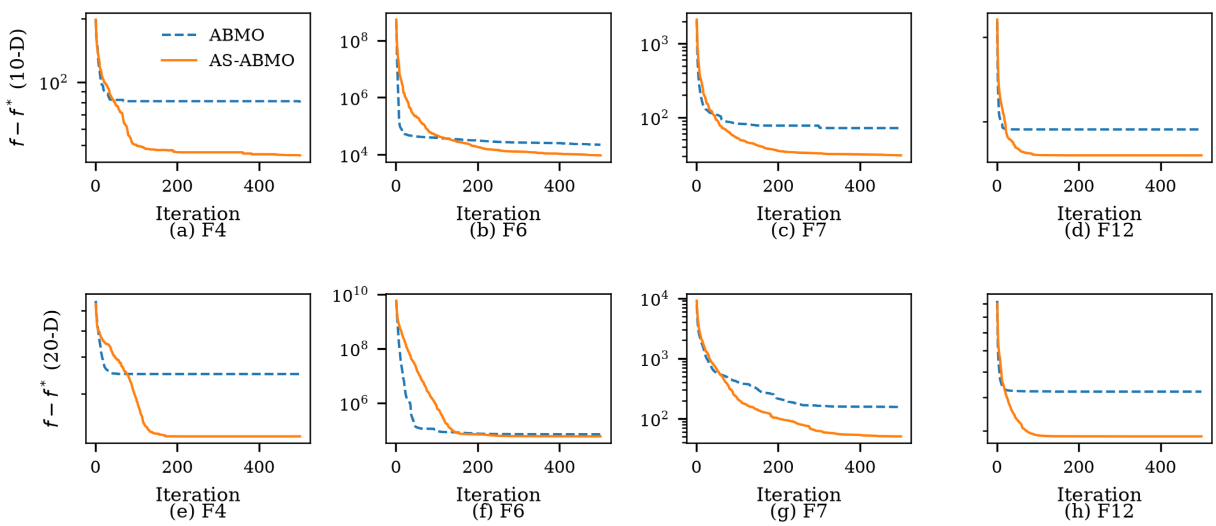

Figure 4: Mean convergence of ABMO and AS-ABMO over 30 runs on F4, F6, F7 and F12 of CEC2022 in $D = 10$ (a–d) and $D = 20$ (e–h); the error $f - f^*$ to the known optimum is shown on a logarithmic scale.

Table 5: CEC2022, $D = 10$: mean and standard deviation of the final fitness over 30 runs; best mean per function in bold.

		ABMO	AS-ABMO	RUN	GWO	WOA	HHO	SCA	L-SHADE
F1	Mean	300.00	300.00	956.84	426.81	3593.73	1398.03	1395.19	**300.00**
	Std	1.03E-13	5.28E-14	364.62	51.20	3144.89	795.13	499.52	0.00E+00
F2	Mean	413.74	407.51	458.77	436.87	445.73	517.98	526.87	**405.62**
	Std	18.77	12.16	55.81	24.44	51.71	89.71	27.02	3.42
F3	Mean	600.00	**600.00**	600.02	600.01	600.00	600.01	600.05	600.00
	Std	3.99E-03	1.10E-13	8.21E-03	2.16E-03	3.48E-03	5.43E-03	0.02	0.00E+00
F4	Mean	879.79	844.90	840.56	841.25	896.79	862.30	861.90	**814.53**
	Std	35.82	17.43	10.42	10.19	34.86	27.87	14.00	4.11
F5	Mean	902.15	900.22	901.01	900.14	902.78	901.47	900.46	**900.00**
	Std	1.52	0.45	0.43	0.21	1.52	0.91	0.09	0.00E+00
F6	Mean	24123.91	11281.25	36246.00	1.22E+06	26016.08	40813.93	3.92E+06	**1801.30**
	Std	14827.30	7877.37	65955.35	9.75E+05	14690.94	24071.36	2.41E+06	0.97
F7	Mean	2071.76	2030.70	2107.25	2071.81	2267.14	2240.77	2107.40	**2012.07**
	Std	49.78	25.82	42.79	34.84	181.43	164.09	18.71	6.31
F8	Mean	2314.48	2221.70	2792.92	2395.12	4557.12	3497.15	2664.26	**2213.10**
	Std	187.72	5.49	554.22	180.07	1708.99	1104.85	245.74	8.49
F9	Mean	2668.87	2603.91	2539.99	2620.18	2741.30	2853.09	2531.68	**2431.67**
	Std	94.46	138.30	187.46	128.23	172.92	180.79	88.41	176.00
F10	Mean	2913.87	2769.79	2689.60	2734.93	2878.68	2775.01	2646.55	**2634.39**
	Std	457.77	319.00	58.11	212.94	432.84	305.54	24.26	33.39
F11	Mean	2782.44	2836.58	2640.36	2782.38	2702.03	2726.99	2632.12	**2629.20**
	Std	353.79	392.69	158.88	354.14	270.86	278.85	5.21	159.95
F12	Mean	2892.23	2869.72	2871.61	2870.15	3012.05	3011.49	2916.90	**2865.24**
	Std	56.43	9.89	1.95	5.84	122.86	92.31	13.91	0.98
Friedman rank		4.79	3.21	4.50	4.33	6.58	6.42	5.17	1.00

Table 6: CEC2022, $D = 20$: mean and standard deviation of the final fitness over 30 runs; best mean per function in bold.

		ABMO	AS-ABMO	RUN	GWO	WOA	HHO	SCA	L-SHADE
F1	Mean	307.13	300.01	10598.42	1875.68	11266.64	11593.19	13875.99	**300.00**
	Std	26.19	0.05	2859.29	595.05	5785.51	3434.02	2828.66	8.94E-13
F2	Mean	461.46	459.32	869.03	516.48	692.31	837.94	1036.48	**449.73**
	Std	26.34	18.91	135.76	21.88	131.72	158.55	184.15	15.11
F3	Mean	600.01	600.00	600.35	600.08	600.24	600.32	600.54	**600.00**
	Std	0.02	1.05E-12	0.12	0.01	0.15	0.15	0.11	5.17E-14
F4	Mean	1048.23	925.20	1001.91	1010.65	1178.17	1066.71	1087.72	**873.35**
	Std	63.18	42.39	30.33	29.76	67.57	55.76	36.12	14.50
F5	Mean	905.57	902.14	903.94	901.56	908.40	904.09	904.94	**900.11**
	Std	2.86	1.75	0.56	0.97	2.93	1.83	0.94	0.28
F6	Mean	71718.26	60629.74	1.39E+08	4.13E+07	7.72E+06	6.21E+07	5.51E+08	**29963.45**
	Std	34373.69	29858.89	1.19E+08	1.76E+07	3.31E+07	1.04E+08	2.40E+08	10368.53
F7	Mean	2155.89	2050.72	2724.58	2264.84	3422.70	3992.85	3098.09	**2030.44**
	Std	218.06	30.40	348.85	63.34	623.45	1111.28	355.06	3.83
F8	Mean	3381.22	3131.58	4669.60	11868.05	5752.83	1.30E+05	3.42E+07	**2229.98**
	Std	1572.94	1175.73	1655.91	17028.23	1790.16	6.83E+05	6.35E+07	4.76
F9	Mean	2669.82	2643.12	2971.41	2708.15	2906.11	3348.47	3500.81	**2635.64**
	Std	49.97	21.14	112.74	35.17	167.28	587.09	194.72	1.15E-04
F10	Mean	4477.22	4234.11	3903.62	4950.20	5390.58	5632.70	**2964.94**	3025.37
	Std	836.67	745.16	1790.30	1835.07	1369.27	962.18	132.06	493.05
F11	Mean	2884.78	2776.01	2836.51	2637.54	3289.13	3518.83	2956.74	**2600.00**
	Std	535.97	478.22	314.07	11.11	810.13	1091.67	157.69	1.35E-08
F12	Mean	3121.04	2986.18	3011.64	2986.57	3271.59	3490.40	3360.06	**2943.86**
	Std	140.97	38.79	15.67	24.26	147.47	312.55	58.43	10.99
Friedman rank		4.00	2.33	4.92	4.00	6.08	6.83	6.75	1.08

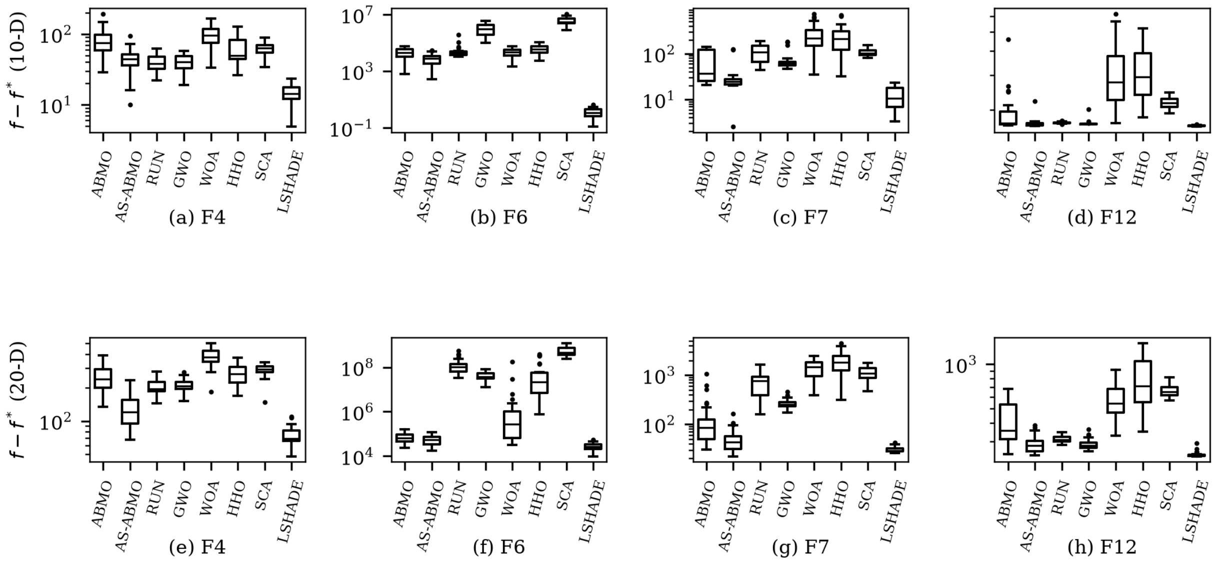

Figure 5: Distribution of the final error $f - f^*$ over 30 runs for the eight algorithms on F4, F6, F7 and F12 in $D = 10$ (a–d) and $D = 20$ (e–h).

4.6 Behaviour of the step scale

Figure 6 shows the mean trajectory of s_t on the same four functions. In all cases the scale rises to its upper bound within the first 20 iterations, when almost every move is accepted, and then follows the difficulty of the landscape. On the unimodal-like F4 and on the composition function F12 it decreases monotonically to the lower bound as the population converges, which turns the late phase into a fine local search. On the hybrid functions F6 and F7 the scale falls, recovers and falls again, which corresponds to escapes from local optima followed by renewed convergence. The trajectories in $D = 20$ stay at the upper bound longer than in $D = 10$, consistent with the larger gains observed in the higher dimension.

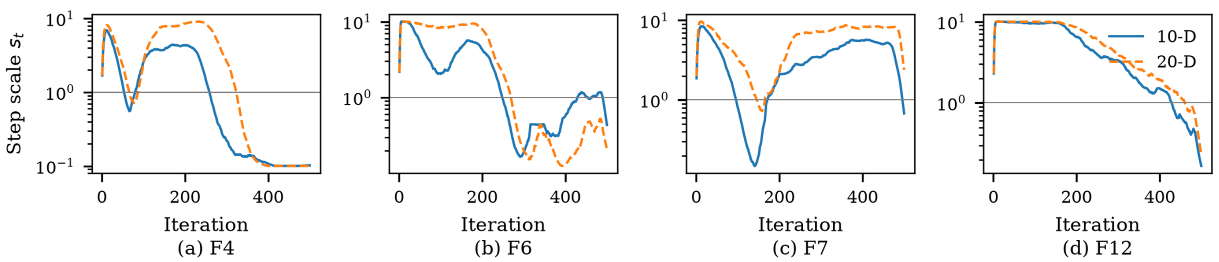

Figure 6: Mean trajectory of the step scale s_t of AS-ABMO over 30 runs on F4, F6, F7 and F12 of CEC2022 in $D = 10$ and $D = 20$; the horizontal line marks $s_t = 1$, the step of the base algorithm.

4.7 Computational cost

Table 7 lists the wall time per run for the same budget of 15 030 evaluations. AS-ABMO and ABMO require the same time within the measurement noise (1.48 s against 1.50 s in $D = 10$), so that the two modifications are free in practice. RUN, SCA and L-SHADE are 1.5 to 2 times slower in the implementation used here.

Table 7: Mean wall time per run in seconds (twelve functions, five seeds each) for 15 030 function evaluations.

	ABMO	AS-ABMO	RUN	GWO	WOA	HHO	SCA	L-SHADE
$D = 10$	1.50	1.48	2.32	1.64	1.50	1.39	2.19	2.40
$D = 20$	1.88	1.89	2.81	2.11	1.96	1.82	3.21	2.83

5 Constrained engineering design problems

Seven classical constrained design problems are used: the three-bar truss, the tension/compression spring, the pressure vessel, the welded beam, the speed reducer, the cantilever beam and the multiple disk clutch brake. The formulations follow the standard references [6, 20, 22] and are reproduced in the supplementary material. The speed reducer has one integer variable (number of teeth) and the clutch brake one (number of friction surfaces); these are rounded after each move. For the welded beam the polar moment of inertia $J = 2\sqrt{2}\,hl\,[l^2/4 + ((h+t)/2)^2]$ and the deflection $\delta = 4PL^3/(Et^3b)$ are used; the formulation-dependent optimum of this problem explains the range of values reported in the literature. For the clutch brake the unit convention of the reference implementation is followed, in which the friction torque is evaluated in N mm.

To anchor the comparison, the exact optimum of every problem was computed by sequential quadratic programming from 100 random starts, with the integer variables enumerated over their ranges; the results are listed in the first column of Table 8. The metaheuristics use $N = 30$, $T = 500$, 30 runs and the feasibility rule of Section 2.3, so that no penalty coefficients enter the comparison. GWO, WOA, SCA and HHO were re-implemented in the same framework so that the constraint handling is identical for all algorithms.

The medians of ABMO and AS-ABMO coincide with the exact optimum on the truss, welded beam, cantilever and clutch brake problems, and are within 0.05% of it on the spring. On the pressure vessel both variants find the continuous optimum 5885.33 in their best run but the median remains 2 to 3% above it, which reflects the narrow feasible corridor of this problem at the fixed budget. On the speed reducer the median of AS-ABMO (2999.1) is 0.16% above the exact value that ABMO attains; this is the only problem on which the modification is significantly worse ($p < 0.001$), and it is the problem with the largest number of active constraints and an integer variable, for which the added dispersion of the individual-aware predictor is not helpful. On no problem does either variant end in an infeasible design. The classical competitors are clearly behind: GWO, WOA, SCA and HHO reach the exact optimum only on the clutch brake and, for WOA, the truss, and their means on the pressure vessel and the welded beam are 10 to 80% above the optimum. The Friedman ranks by mean over the seven problems are ABMO 1.64, AS-ABMO 2.07, WOA 3.57, GWO 4.14, SCA 4.29 and HHO 5.29. The conclusion for design practice is that both ABMO variants solve the classical low-dimensional problems to their exact optima; the advantage of AS-ABMO documented in Section 4 lies in higher-dimensional and multimodal landscapes and does not translate into a further gain on problems that the base algorithm already solves.

Table 8: Constrained design problems, 30 runs, feasibility rule for all algorithms: exact optimum (multistart SQP), median (mean) of ABMO and AS-ABMO, and mean of the classical competitors. No run of any algorithm ended infeasible.

Problem	Exact	ABMO	AS-ABMO	GWO	WOA	SCA	HHO
Three-bar truss	263.8958	263.896 (263.896)	263.896 (263.896)	282.84	265.18	264.13	264.25
Spring	0.012665	0.012719 (0.012865)	0.012719 (0.012810)	0.01367	0.01353	0.01311	0.01371
Pressure vessel	5885.33	6003.2 (6183.5)	6074.0 (6329.5)	6520.6	6631.8	7220.0	66596
Welded beam	1.695247	1.69525 (1.69529)	1.69525 (1.69533)	1.7079	1.7180	1.8791	3.0742
Speed reducer	2994.471	2994.47 (3075.9)	2999.14 (3155.5)	3023.3	3009.8	3138.6	4264.8
Cantilever beam	1.339956	1.33996 (1.33997)	1.33997 (1.33997)	1.3414	1.3408	1.4135	1.4841
Clutch brake	0.235242	0.235242 (0.235242)	0.235242 (0.235242)	0.23695	0.23524	0.23625	0.23524

6 Application: lightweight design of an automotive component

6.1 Component, design variables and load cases

The component is a steel [bracket / plate-type automotive part, to be named] of S235 structural steel (yield strength 235 MPa; E, ν and ρ in Table 10) whose mass is reduced by material removal. Eight geometric parameters of the cut-outs are the design variables (Table 9): the length and depth of a cut along the right edge (x_1, x_2), the length and depth of a corner cut-out (x_3, x_4), the two dimensions of a central cocoon-shaped cut (x_5, x_6) and the height and width of a central rectangular opening (x_7, x_8). Two design rules limit the combined cut lengths, $x_2 + x_4 \leq 80$ mm and $x_1 + x_3 \leq 260$ mm. The finite element model, built and solved in [Ansys Mechanical, version], uses [element type, mean element size]; a mesh convergence study on the baseline design with three mesh densities changed the maximum von Mises stress by less than [x]% (Table 10). The boundary conditions and load surfaces are shown in Fig. ??. Two load cases are considered. Load case 1 is the bending load with a fatigue requirement used in a previous study of the same component [previous component study, to be cited when published]; it serves here to test the surrogate pipeline and to illustrate the effect of an inactive constraint. Load case 2, [torsion / combined bending and torsion], is the design case of this paper; its load level was chosen so that the stress constraint is active in the region of interest.

6.2 Design of experiments and surrogate models

A Latin hypercube design of 131 points over the variable bounds was evaluated by finite element analysis for both load cases (the same input matrix is used, so the two response sets are directly comparable). Gaussian process regression (GPR) surrogates were fitted to the mass, the maximum von Mises stress, the maximum total deformation and [the fatigue life / the first natural frequency], using a Matérn 5/2 kernel with one length scale per variable and an additive white noise term; the fatigue life is modelled in $\log_{10}$. Inputs are scaled to the unit cube and outputs are standardized; hyperparameters are obtained by marginal likelihood maximization with five restarts. Five-fold cross-validation (Table 11, Fig. 7) gives coefficients of determination of [0.975 / 0.949 / 0.984 / 0.931 for load case 1: mass, stress, deformation, $\log_{10}$ life] and [values for load case 2].

6.3 Optimization and verification

The design problem is

$$\min_{\mathbf{x}} \ \hat{m}(\mathbf{x}) \quad \text{s.t.} \quad \hat{\sigma}(\mathbf{x}) \leq \sigma_y/\text{SF}, \ \ \hat{\delta}(\mathbf{x}) \leq \delta_{\text{allow}}, \ \ [\hat{N}(\mathbf{x}) \geq 10^7 \text{ or } \hat{f}_1(\mathbf{x}) \geq f_{\min}], \ \ x_2+x_4 \leq 80, \ \ x_1+x_3 \leq 260, \tag{11}$$

with SF = 1.2, where hats denote surrogate predictions. AS-ABMO is run on the surrogates with the settings of Section 4 ($N = 30$, $T = 500$, feasibility rule) from 30 seeds; one run takes about one second. The surrogate optima are then verified by finite element analysis, together with three exploratory feasible designs selected by farthest-point sampling within 3% of the predicted optimum mass so that the surrogates are corrected where the active constraint is located. The verified designs are added to the training set, the surrogates are refitted and the optimization is repeated; two such infill rounds were sufficient for the predicted and verified responses of the optimum to agree within [x]%.

For load case 1 the 30 runs converge to two designs of practically equal predicted mass (1.138 kg) that differ only in the central cut ($x_6 = 35$, $x_8 = 100$ against $x_6 = 95$, $x_8 = 88.5$ mm); both lie on the fatigue constraint, which is the only active constraint since the stress remains at 124–131 MPa for every design of the DoE, far below the allowable 196 MPa. The reference design of that study, with all cut-outs at their upper bounds, violates the fatigue requirement (6.4×10^6 cycles), so that the feasible optimum is 2.1% heavier than the reference; the problem therefore has no lightweighting margin under load case 1, which motivated load case 2.

For load case 2 [results: Table 12 compares the baseline and the optimized design (mass, stress, deformation, [life/frequency], constraint margins), predicted against verified values; Fig. **??** shows the two geometries and the stress fields]. The optimized design reduces the mass by [x]% relative to the [feasible] baseline while satisfying all constraints in the verification analysis with a margin of [x]% on the stress. For comparison, ABMO and [GWO] were run on the same surrogates with the same budget; [result of the comparison].

Table 9: Design variables of the component: code, description, bounds (mm) and baseline value.

Variable	Description	Lower	Upper	Baseline
x_1	horizontal cut length along the right edge	63.01	260	260
x_2	vertical cut-out depth at the right edge	3.01	37.5	37.5
x_3	horizontal cut-out length at the lower right corner	20.01	60	60
x_4	vertical cut-out depth at the lower right corner	13.51	42.5	42.5
x_5	vertical size of the central cocoon-shaped cut	5	25	25
x_6	horizontal size of the central cocoon-shaped cut	5	95	95
x_7	height of the central rectangular opening	7	80	80
x_8	width of the central rectangular opening	7	100	100

Table 10: Finite element setup and material data. [To be completed from the FEA documentation.]

Solver, version	[Ansys Mechanical 20xx Rx]
Element type, size	[SOLID187, x mm]
Mesh convergence (3 densities, baseline)	[$\Delta\sigma_{\max} < x\%$]
Material	S235, E = [210] GPa, ν = [0.3], ρ = [7850] kg/m^3, $\sigma_y = 235$ MPa
Load case 1	[bending, F = ..., fatigue: ...]
Load case 2	[torsion / combined, T = ...]
Allowables	$\sigma_y/1.2 = 195.8$ MPa; δ_{allow} = [..] mm; [$N \geq 10^7$ / $f_1 \geq$ .. Hz]

Table 11: Five-fold cross-validation of the GPR surrogates (R^2 and RMSE in the units of the response; fatigue life in $\log_{10}$ cycles).

	Load case 1		Load case 2	
Response	R^2	RMSE	R^2	RMSE
Mass (kg)	0.975	0.0115	[..]	[..]
Max. von Mises stress (MPa)	0.949	0.245	[..]	[..]
Max. deformation (mm)	0.984	0.00019	[..]	[..]
Fatigue life ($\log_{10}$ cycles) / [frequency]	0.931	0.0185	[..]	[..]

Table 12: Baseline and optimized design under load case 2: predicted (surrogate) and verified (FEA) responses. [To be completed.]

[TABLE PLACEHOLDER]

Fig. E placeholder: fig/figE_gpr_cv.png

Figure 7: Five-fold cross-validated GPR predictions against finite element results for load case 1: (a) mass, (b) maximum von Mises stress, (c) maximum deformation, (d) $\log_{10}$ fatigue life.

7 Discussion and managerial implications

Three points of the results deserve comment. First, the reproduction study shows that a faithful re-implementation of a recently published optimizer can differ from the published numbers by more than the run-to-run variation on individual functions, even when the overall picture is reproduced; releasing code with such papers, as done here, is the only way to remove this ambiguity. Second, the collinearity of the ABMO stages is a property of the algorithm rather than of its implementation, and it limits what can be borrowed from numerical integration: the predictor corrector structure provides a convenient parameterization of the move, but not an error estimate. Feedback from accepted moves, a device that predates most metaheuristics, is what makes the step adaptive. Third, the gains of AS-ABMO are concentrated in multimodal and higher-dimensional landscapes and do not appear on the classical low-dimensional design problems, which both variants solve exactly. For a design engineer this means that the choice between the two is immaterial for textbook problems with a handful of variables, while for surrogate-assisted component design with ten or more shape variables and a rugged response the adaptive variant is the safer default at no additional cost. The exact optima reported in Table 8 also give practitioners a reference against which any optimizer can be checked before it is applied to a component.

8 Conclusions

This paper presented AS-ABMO, an adaptive-step variant of the Adams-Bashforth-Moulton optimizer. An analysis of the base algorithm showed that its predictor corrector stages are collinear, so that the step size cannot be controlled by the error estimate of the underlying integration method; a success-rate controller and an individual-aware predictor were introduced instead, and an ablation study showed that only their combination improves the base algorithm, on eight of twelve CEC2022 functions in ten dimensions and five of twelve in twenty dimensions, with no significant loss and no measurable cost. Under an equal evaluation budget AS-ABMO ranked second among eight optimizers behind L-SHADE and ahead of RUN, GWO, WOA, HHO and SCA. On seven constrained design problems both ABMO variants reach the exact optima computed by sequential quadratic programming, with one exception on the speed reducer where the modification is slightly worse. [Component result sentence.] The limitations of the study are the restriction to CEC2022 and to a single evaluation budget, the absence of NRBO and QIO from the competitor set because no verified implementations were available, and the single component application. Future work will extend the controller to the balance factor E_t, examine multi-objective and discrete variants, and apply the surrogate-assisted workflow to further components of the vehicle structure.

Data availability

The Python implementation of ABMO and AS-ABMO, the re-implemented competitors, all raw results (per-run CSV files), the figure data and the exact-optimum computations are available at https://github.com/KMDR82/ASABMO.

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